

Qingyang Xiong

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2 Choi Sha Street, Ma On Shan, Sha Tin, New Territories

EDUCATION

- Nanjing University of Information Science & Technology** 2018-2022
B.S. in Atmospheric Sciences **Advisor:** Juan Li GPA: 3.0/4.0
- Second Institute of Oceanography, MNR** 2022-2025
M.S. in Physical Oceanography **Advisor:** Qiaoyan Wu GPA: 4.0/4.0
- City University of Hong Kong** 2025-Present
Ph.D. in Engineering **Advisor:** Ping Lu

RESEARCH PROJECTS

- Evaluation of CMIP6 Models' Performance on Simulating the Asian-Australian Monsoon.** 2021-2022
Undergraduate thesis
 - Keyword:** Python, SEOF, CMIP6
 - Evaluated the performance of 13 climate models and their Multi-Model Ensemble (MME) in simulating the climatology of the Asian-Australian Monsoon.
- The Relationship between Maximum Intensification Rate and Lifetime Maximum Intensity of TC.** 2023-2025
Main research topic of postgraduate
 - Keyword:** Carnot Heat Engine Theory, Differential Equation, WRF Ideal Experiment, Python
 - Based on the Carnot engine theory and the intensification rate (IR) equation of tropical cyclone (TC). Primary focus on how the maximum intensification rate (MIR) of TC influences its lifetime maximum intensity (LMI). The discussion examines the timing and magnitude relationships. Validation of the conclusions is conducted best-track datasets and full-physics numerical simulation.
- Modulation of Concentric Eyewall Storms by Climate Modes.** 2025-2026
First Project during PhD study
 - Keyword:** Tropical Cyclones, Concentric Eyewall, Climate Mode
 - Using 30 years of global concentric eyewall (CE) observations (1991–2020), this study examined how ENSO, the Indian Ocean Basin Mode (IOBM), and Atlantic Niño/Niña (ATLM) modulate CEs across the Western North Pacific, Eastern Pacific, North Atlantic, and Indian Ocean basins, and analyzed the associated large-scale environments (SST, MPI, humidity, vertical wind shear, genesis location, fetch distance, and exposure).
- Toward a Theoretical Radial Wind Profile Model for Concentric Eyewall.** 2026-
Second Project during PhD study
 - Keyword:** Radial wind profile, Concentric Eyewall, SAR Observation
 - Using high-resolution SAR observations, this project decomposes a concentric eyewall tropical cyclone into two embedded single-eyewall-like substructures, applies appropriate single-eyewall models within their defensible radial ranges, and reconnects the pieces at dynamically meaningful anchor points to build a physically interpretable theoretical radial wind framework.

SKILLS & INTERESTS

Programming: Python, L^AT_EX, Matlab, Fortran

Skills: Dynamic Meteorology, tcpyPI, WRF, Overleaf

Field of Interest: Atmospheric Dynamics, Tropical Cyclone Dynamics, Data science

Language Proficiency: TOEFL iBT: 81

Hobbies: Basketball, Physics, Reading

ACHIEVEMENTS

- Full marks (4.0/4.0) in Dynamic Meteorology at undergraduate (92/100) and graduate level (100/100)**
- First class scholarship for graduate students, SIO** 2023
- Second class scholarship for graduate students, SIO** 2024
- Oral presentation at Xiamen Symposium on Marine Environmental Sciences, Xiamen** 2025
- Excellent oral presentation at The 2nd Future Atmospheric Science Forum, Suzhou** 2026

- **The Intrinsic Timing Relationship between Tropical Cyclone Maximum Intensification Rate and Lifetime Maximum Intensity.**

Qingyang Xiong & Qiaoyan Wu. 2025 *Geophysical Research Letters*, *Accepted*

<https://doi.org/10.1029/2024GL113158>

Abstract. The intensification rate (IR) equation, derived from a time-dependent simplified energetically based dynamical system (EBDS), links tropical cyclone (TC) intensity, IR, and their evolutions, suggesting a connection between the timing of TC lifetime maximum intensity (LMI) and the maximum intensification rate (MIR). This study demonstrates that the IR equation predicts MIR typically preceding LMI within 36 hours, as confirmed by numerical simulation and analysis of best track data. It was observed that the majority of global TCs reached LMIs within 36 hours after MIR from 1982 to 2023. This consistency between observational data, numerical simulations and analytical results highlights the EBDS-based IR equation's ability in capturing the timing relationship between MIR and LMI. This intrinsic link highlights the importance of understanding the timing of MIR in the evolution of TCs.